

Analysis of clinical parameters that contribute to the misdiagnosis of biliary atresia.

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Biliary atresia (BA) is one of the most common and perplexing causes of neonatal cholestasis. Each year many cases of neonatal cholestasis are misdiagnosed as BA, resulting in unnecessary exploratory surgery. Therefore, the aim of our study was to analyze the clinical features and parameters that contribute to the misdiagnosis of BA. We used a retrospective study to compare BA and similar neonatal cholestatic diseases, which were confirmed by intraoperative cholangiography. Six hundred and two infants that were suspected to have BA were recruited for the study. All cases were divided into a non-BA group and a BA group according to intraoperative cholangiography. In addition, each group was divided into three subgroups according to the patients age at surgery (group i, 90 d). The annual misdiagnosis rate of non-BA patients was calculated. Age at onset of jaundice and the liver function and ultrasound results were compared between the two groups and subgroups. Moreover, the positive predictive value and false positive rate of hepatobiliary scintigraphy in the diagnosis of BA were calculated. Finally, the disease spectrum of the non-BA group was analyzed. Of the 602 cases, 83 patients were diagnosed as non-BA. The remaining 519 cases were confirmed to have BA. There was no significant decline in the misdiagnosis rate of suspected BA cases by year. The age at onset of jaundice, total bilirubin (TBIL), direct bilirubin (DBIL), DBIL/TBIL and alanine aminotransferase (ALT) values before the exploratory operation showed no statistically significant difference ($P > 0.05$) in the non-BA versus BA groups. However, the mean level of gamma-glutamyl transpeptidase (γ -GT) was 263.2 mmol/l in the non-BA group and 902.7 mmol/l in the BA group ($P < 0.01$). The length of the liver below the ribs was detected with ultrasound and found to be smaller in the non-BA group than that of the BA group ($2.99 \text{ cm} \pm 1.62$ vs. $3.61 \text{ cm} \pm 1.26$, respectively; $P < 0.05$). Among the 498 infants who received hepatobiliary scintigraphy examination, the false positive rate was 13.3% (66/498) and the positive predictive value was 86.7% (432/498). In the non-BA group, 58 infants suffered from hepatitis syndrome, 16 cases were biliary dysplasia, 5 cases were TPN related cholestasis, 2 cases were bile duct perforation and 2 were bile-plug syndrome. The similarity of liver function tests and excessive dependence on hepatobiliary scintigraphy may contribute to the misdiagnosis of infants with jaundice. The age at onset of jaundice, the level of γ -GT and the liver length below the ribs may be helpful in the differential diagnosis of jaundice in infants.

The Assessment and Management of Biliary Atresia in Hawai'i, 2009-2023.

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Although biliary atresia (BA) is a rare neonatal disorder, it remains the leading cause of pediatric end-stage liver disease. Early diagnosis of BA and treatment with the Kasai procedure can significantly reduce the need for pediatric liver transplant. Current data suggests that performing the Kasai procedure at 30-45 days of life is associated with longer native liver survival rates and reduction of the need for liver transplant. The incidence rate of BA in the state of Hawai'i is nearly double the incidence rate in the continental US. International studies have demonstrated that screening programs for BA reduce the age at diagnosis and treatment. However, there has been no statewide analysis on the ages at diagnosis or at Kasai, nor does a statewide screening program for BA exist. The purpose of this study is to review the age of diagnosis and treatment of BA to determine if the current practice in Hawai'i is in line with the published data. A retrospective chart review of all patients diagnosed with BA at the state's primary children's hospital was performed (2009-2023) and 19 patients who underwent the Kasai procedure were identified. The mean age at diagnosis is 71.4 days ($n=19$) and the mean age at Kasai procedure is 72.0

days (n=19). Both the average age at diagnosis and treatment for BA in Hawai'i is significantly higher than published data suggesting best outcomes at 30-45 days of life. This review suggests that the implementation of a statewide screening program for BA in Hawai'i is warranted.

Biliary atresia: pathogenesis and treatment.

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Biliary atresia is a disorder of infants in which there is obliteration or discontinuity of the extrahepatic biliary system, resulting in obstruction of bile flow. Untreated, the resulting cholestasis leads to progressive conjugated hyperbilirubinemia, cirrhosis, and hepatic failure. Biliary atresia has an incidence of approximately one in 10,000 live births worldwide. Evidence to date supports a number of pathogenic mechanisms for the development of biliary atresia. An infectious cause, such as by a virus, would seem most plausible in many cases. The clinical observation that biliary atresia is rarely encountered in premature infants would support an agent acting late in gestation. However, no infectious or toxic agent has been conclusively implicated in biliary atresia. Genetic mechanisms likely play important roles, even regarding susceptibility to other specific causes, but no gene whose altered function would result in obstruction or atresia of the biliary tree has been identified. The variety of clinical presentations support the notion that the proposed mechanisms are not mutually exclusive but may play roles individually or in combination in certain patients. Biliary atresia, when untreated, is fatal within 2 years, with a median survival of 8 months. The natural history of biliary atresia has been favorably altered by the Kasai portoenterostomy. Approximately 25 to 35% of patients who undergo a Kasai portoenterostomy will survive more than 10 years without liver transplantation. One third of the patients drain bile but develop complications of cirrhosis and require liver transplantation before age 10. For the remaining one third of patients, bile flow is inadequate following portoenterostomy and the children develop progressive fibrosis and cirrhosis. The portoenterostomy should be done before there is irreversible sclerosis of the intrahepatic bile ducts. Consequently, a prompt evaluation is indicated for any infant older than 14 days with jaundice to determine if conjugated hyperbilirubinemia is present. If infectious, metabolic, endocrine disorders are unlikely and if the child has findings consistent with biliary atresia, then exploratory laparotomy and intraoperative cholangiogram should be done expeditiously by a surgeon who has experience doing the Kasai portoenterostomy. Biliary atresia represents the most common indication for pediatric liver transplantation, representing more than 50% of cases in most series. Transplantation is indicated when symptoms of end stage liver disease occur, including recurrent cholangitis, progressive jaundice, portal hypertension complications, ascites, decreased synthetic function, and growth/nutritional failure.

Biliary atresia.

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Biliary atresia, a progressive obliterative process involving the bile ducts, has its onset in the newborn period. It is characterized by worsening cholestasis, hepatic fibrosis, and cirrhosis, which lead to portal hypertension and a decline in hepatic synthetic function. Untreated, the outcome is uniformly fatal. The two major milestones toward improved treatment of this disease have been the Kasai portoenterostomy and orthotopic liver transplantation. There has been discussion regarding transplantation as primary therapy, but portoenterostomy remains the standard of care as first-line intervention. Hepatic

transplantation, done more frequently for biliary atresia than for any other cause of liver failure in the pediatric population, offers improved survival and quality of life to those for whom the Kasai operation fails. The etiology of biliary atresia remains poorly understood. Working toward a better understanding of this disease, recent investigations target more precise characterization of the hepatic pathology and seek to identify possible causative agents and predictors of favorable outcome. Recent advances in the understanding of biliary atresia published between December 1995 and November 1996 are the focus of this review.

[Biliary atresia: a severe illness].

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Biliary atresia is a serious disease of unknown cause, affecting newborns. An inflammation and progressive destruction of the bile ducts lead to jaundice, dark urines, and acholia, between the second and sixth weeks of life. Neonatal cholestasis could be due to several different diseases, thus a diagnosis of biliary atresia and early derivation for surgical treatment are necessary to allow a restoration of the bile flow. Eighty percent of the children normalize serum bilirubin after the portoenterostomy (Kasai operation), if they are operated before their 45 days of life. When Kasai operation fails, a liver transplantation is the only possibility. Biliary atresia must be diagnosed before the first month of life and must be considered as a surgical emergency.

Biliary atresia: an overview.

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Biliary Atresia is the most common cause of persistently direct (conjugated) hyperbilirubinemia in the first three months of life. It is a progressive inflammatory obliteration of the extra- and intrahepatic bile ducts whose pathogenesis remains speculative. This review updates the history, etio-pathological considerations, clinical manifestations, methods of diagnosis, surgical management, and results of this terrible disease in young infants. We will emphasize that persistent jaundice in the newborn period must be managed urgently, a diagnosis should be established early in life, and Kasai portoenterostomy offered to those infants with Biliary Atresia before their eighth week of life. This will allow more than one-third of children to survive. Hepatic transplantation is reserved for those children with failed portoenterostomy, progressive liver failure or late-referral to surgery.

Etiopathogenesis of biliary atresia.

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Biliary atresia, a progressive sclerosis of the extrahepatic biliary tree that occurs only within the first 3 months of life, is one of the most common causes of neonatal cholestasis and accounts for over half of children who undergo liver transplantation. In biliary atresia, a number of prenatal or perinatal insults to the biliary tree appear to culminate in complete obliteration of the lumen of the extrahepatic biliary tree and continued injury and sclerosis of intrahepatic bile ducts, even after portoenterostomy is successful. A

minority of cases of biliary atresia may be caused by defects in morphogenesis of the bile ducts. Potential etiologies for the more common perinatal form of biliary atresia include viral infections, immune-mediated bile duct injury, and autoimmune disease involving the bile ducts. Two viruses, reovirus and rotavirus, have received increasing attention as possible inciters of an immune-mediated injury to the biliary tree. Fas ligand upregulation and apoptosis of bile duct epithelia have been demonstrated in human specimens, as well as T-lymphocyte and macrophage activation in portal tracts. An experimental model using rotavirus infection in newborn mice has been useful in characterizing the mechanisms underlying bile duct injury. It is proposed that virally induced neoantigens displayed on biliary epithelium may play a role in initiating the immune processes involved in destruction of the extrahepatic bile duct and ongoing intrahepatic ductal injury in the perinatal form of biliary atresia. The short window of time after birth during which this disease presents suggests that immaturity of the neonatal immune system and genetic susceptibility also may be key factors. Delineation of the mechanisms underlying bile duct injury will be essential to the development of new potential therapies for this important pediatric disorder.

Biliary Atresia: Clinical Phenotypes and Aetiological Heterogeneity.

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Biliary atresia (BA) is an obliterative condition of the biliary tract that presents with persistent jaundice and pale stools typically in the first few weeks of life. While this phenotypic signature may be broadly similar by the time of presentation, it is likely that this is only the final common pathway with a number of possible preceding causative factors and disparate pathogenic mechanisms-i.e., aetiological heterogeneity. Certainly, there are distinguishable variants which suggest a higher degree of aetiological homogeneity such as the syndromic variants of biliary atresia splenic malformation or cat-eye syndrome, which implicate an early developmental mechanism. In others, the presence of synchronous viral infection also make this plausible as an aetiological agent though it is likely that disease onset is from the perinatal period. In the majority of cases, currently termed isolated BA, there are still too few clues as to aetiology or indeed pathogenesis.

Biliary Atresia in an Infant Presenting With Kabuki Syndrome: An Autopsy Report and Review of the Literature.

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Biliary atresia (BA) is an inflammatory obliterative cholangiopathy which is very common during neonatal and infancy period. We present an autopsy report of a BA in an infant suffering from a genetic syndrome.

Transcriptome profiling of biliary atresia from new born infants by deep sequencing.

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Biliary atresia is the major kind of liver disease that mainly affects the new born infants. The pathological and biological mechanism of biliary atresia is still unclear to date. In this work, we attempt to identify

biliary atresia relevant genes and to get the knowledge of the underlying genetic basis. We collected liver samples from new born infants with biliary atresia and congenital choledochocyst, and the RNA-seq technology was used to performed a transcriptome profiling in order to comprehensively study their expression signatures. We identified 877 differentially expressed genes between samples from biliary atresia and congenital choledochocyst patients in total. Several biological pathways related to the immunity and inflammation response were found to involve in the development of biliary atresia. Our results may helps to better investigate the molecular mechanisms of this disease.
